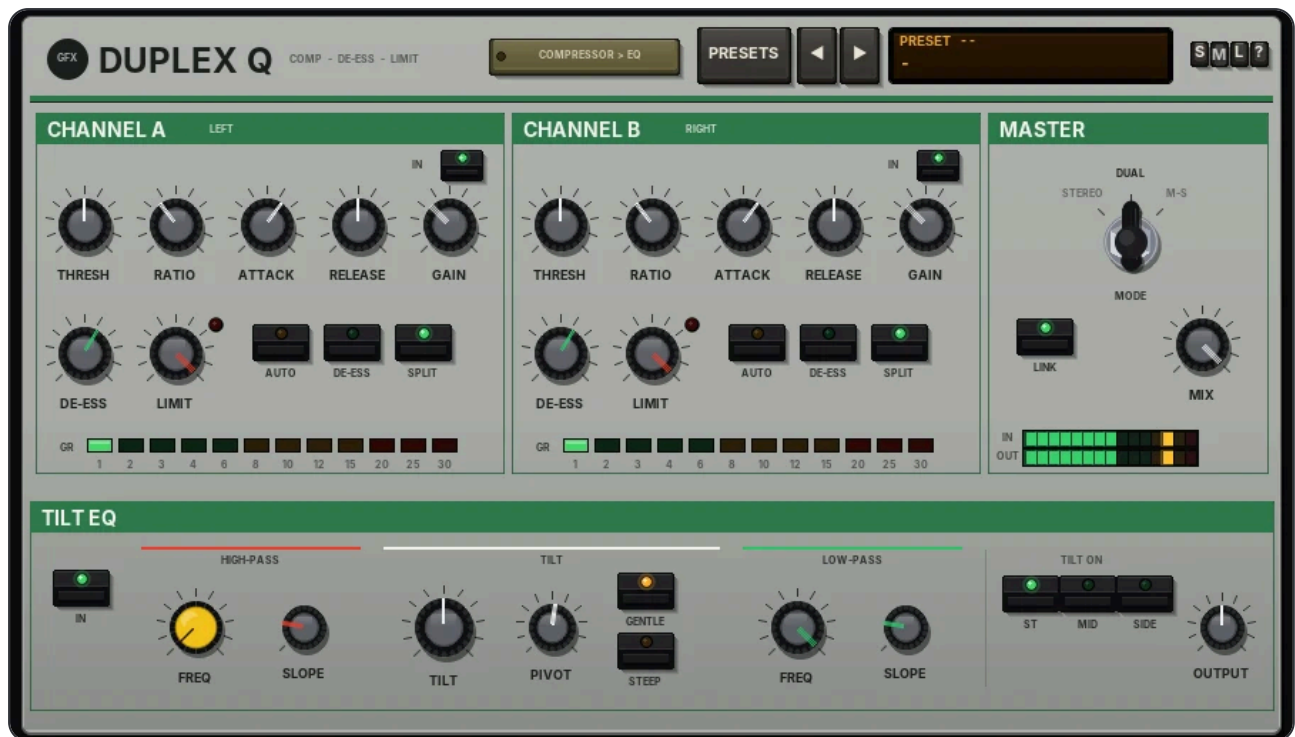


Duplex Q

A two-channel compressor, de-esser and peak limiter with a tilt EQ and filters.



Overview

Duplex Q is a dual-channel dynamics processor in the spirit of the classic 80s British two-channel units, with a tilt EQ and high- and low-pass filters added underneath. Each channel is a clean VCA compressor with a gentle 6 dB knee, a frequency-conscious de-esser and a separate fast peak limiter.

The two channels can run a stereo signal together, work as two independent mono channels (left and right), or work in mid-side, so you can compress the centre of a mix and its edges separately. LINK keeps the stereo image steady and gangs the two channels' controls.

ORDER puts the EQ after the compressors or in front of them. The two units swap places on screen, so the top one is always the one the signal meets first.

Reach for it for

- Vocal compression and de-essing in one place

- Stereo bus glue, or separate control of mid and sides
- Two different mono sources on one plug-in (DUAL)
- A peak safety limiter after the compression
- Quick tonal balancing with TILT and the filters

Controls

CHANNEL A and CHANNEL B

The two channels have the same controls. The label beside each name shows what the channel is working on: BOTH SIDES, LEFT / RIGHT or MID / SIDE. In STEREO mode Channel A runs both sides and Channel B rests (its panel says so).

IN	Switches the channel in or out.
THRESH	The level where compression starts, from -40 to 0 dB (-20 dB by default). The detector measures the average (RMS) level over about 3 ms.
RATIO	From 1:1 up to INF:1 (fully right), 3:1 by default.
ATTACK	From 0.1 to 50 ms (5 ms by default). Shows AUTO when AUTO is on.
RELEASE	From 20 to 2000 ms (200 ms by default). Shows AUTO when AUTO is on.
GAIN	Make-up gain, from -10 to +20 dB.
DE-ESS (knob, green pointer)	The de-ess frequency, from 2 to 12 kHz (6 kHz by default).
LIMIT (red pointer)	A separate fast peak limiter after the compressor, from OFF (fully right, 0 dB) down to -24 dB: the level the peaks are held at. It reacts instantly and recovers in about 50 ms. The red light beside it shows it working.
AUTO	Program-dependent timing: the attack gets faster the further the signal goes over the threshold, and the release

	lets go quickly after short peaks and slowly (up to about a second) after long passages of compression.
DE-ESS (key)	The compressor listens only to what lies above the DE-ESS frequency, so it reacts to sibilance.
SPLIT	With DE-ESS on: the gain reduction is applied only above the de-ess frequency (the split sits at three-quarters of it, so the whole ess band is caught), and the body of the voice is untouched. Off: the whole signal is turned down, as on the original units. On by default.
GR	Twelve lights under each channel showing its gain reduction, from 1 to 30 dB, including the limiter.

MASTER

MODE	STEREO: Channel A runs both sides (Channel B rests). DUAL: A = left, B = right. M-S: A = mid, B = side. Click a name, click the knob to step, or use the mouse wheel.
LINK	In DUAL and M-S, the two channels' detectors share the louder level so the image stays put, and their controls are ganged: turning one moves the other by the same amount, and any offset you set while unlinked is kept. Switches copy across. In STEREO it is always lit, as one detector already runs both sides.
MIX	Parallel compression, shown as WET 0 - 100: blends the dry signal back in. 100 is fully compressed.
IN / OUT meters	Input and output level, with a floating peak hold. Click a meter to clear the hold.

TILT EQ

IN	Switches the filters and the tilt in or out.
HIGH-PASS FREQ	From OFF (fully left, 10 Hz) up to 1 kHz.
HIGH-PASS SLOPE	6, 12, 18, 24, 36 or 48 dB per octave (12 dB by default).

TILT	Tilts the whole spectrum around the pivot, from -6 to +6 dB at each end: positive brightens (more top, less bottom), negative darkens.
PIVOT	The frequency the tilt turns around, from 100 Hz to 5 kHz (800 Hz by default).
GENTLE / STEEP	The shape of the tilt. GENTLE: a broad first-order slope across the whole range. STEEP: a firmer shelf that changes more quickly around the pivot.
LOW-PASS FREQ	From 1 kHz up to OFF (fully right, 22 kHz).
LOW-PASS SLOPE	6, 12, 18, 24, 36 or 48 dB per octave (12 dB by default).
TILT ON: ST / MID / SIDE	Where the tilt works: the whole stereo signal, the mid (centre) only, or the side only. The filters always work on both sides.
OUTPUT	The EQ unit's output level, from -12 to +12 dB.

The header

ORDER	One key: COMPRESSOR > EQ (the default) or EQ > COMPRESSOR. The units swap places on screen. Its amber light shows while you hold it.
PRESETS	Opens the list of factory presets (a click on the display does the same); the ◀ / ▶ keys step through them. Choosing one sets the two channels and the MASTER section; the EQ is left as it is. The last entry resets both channels and the MASTER section to their defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and this manual's address. Click anywhere to close it.

Knob values appear while you touch a knob. A built-in input guard eases very hot sources down before the compressors, so loud sources never overdrive the plugin; it

never boosts quiet ones.

Presets

Gentle Stereo Bus	2:1 at -24 dB with AUTO timing and +3 dB of gain, linked.
Vocal Compressor	3:1 at -28 dB, 5 ms attack, 180 ms release, the limiter at -3 dB.
De-esser (split)	6:1 de-essing above 6.5 kHz with a fast attack, SPLIT on.
Broadcast Voice	4:1 at -26 dB with wideband de-essing at 5 kHz and the limiter at -2 dB.
Dual Mono Pair	DUAL mode, unlinked: two independent 3:1 channels with AUTO timing.
M-S Glue	M-S mode, unlinked: the mid gently at 2.5:1, the sides held harder at 4:1.
M-S Widen + Tame Sides	M-S: light AUTO compression on the mid, the sides lifted 3 dB with de-essing at 7 kHz.
Brickwall Safety	INF:1 at -12 dB with a 0.3 ms attack and the limiter at -1 dB.
Parallel Drums	10:1 at -36 dB with +12 dB of gain, blended in at 40%.
Bass Leveller	4:1 at -24 dB with AUTO timing and the limiter at -2 dB.
Reset all to defaults	Both channels and the MASTER section back to their starting values; the EQ is left as it is.

Tips

- For a vocal, compress on Channel A, then switch on DE-ESS with SPLIT and set the frequency where the esses are loudest - start around 5 - 7 kHz.
- In M-S, compress the side channel a little harder than the mid to steady a wide mix, or lift its GAIN to widen it.

- Use the LIMIT knob as a safety net: set it a few dB under your loudest peaks and watch for the red light.
- Put the EQ first (EQ > COMPRESSOR) when you use the high-pass to stop the low end driving the compressor; put it last for final tonal shaping.

File: gfx-duplex-q.jsfx